Don't Melt Today

A user-generated-content approach to informal cooling refuges and intraday route planning on heat-warning days

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Abstract: The most elaborate, intellectually, of the article's arguments about urban heat response is the claim that local adaptation depends on whether residents can connect cooling centers, shaded places, and isolated facilities into workable daily routes. Using Gelsenkirchen as a case, the study identifies informal cooling refuges from publicly visible user-generated content and translates these traces into an auditable cooling index. Lightweight proxy checks are then used to ask whether the index remains directionally consistent with known cooling conditions such as canopy, vegetation, and water proximity. It is not a question of formal planning or local evidence, but of making the latter supplement the former. Doing so demands an austere discipline of auditability, detachment from anecdote, and some care about what counts as refuge. One has to understand, for one thing, that heat adaptation cannot be reduced to location alone. One needs access, usability, and everyday climate justice.

Keywords: urban heat adaptation; informal cooling refuges; user-generated content; everyday mobility; local planning; climate justice

PART 1 INTRODUCTION

One needs a route perspective. Early in the day, a resident may leave home already under-slept after a hot night, cross an exposed tram platform, run midday errands, and then face a return trip through heat-retaining streets long after peak sunlight has passed. There is a sense in which this whole sequence, too, is what local adaptation has to answer. And through the hot night, the exposed platform, the errands, and the delayed return, we are being prepared - practically, not rhetorically - for cumulative burden. The article takes as its motto for heat risk a whole-day logic rather than the drama of a single emergency. The burden of heat is, of course, most radical when it accumulates through repeated exposure, delayed recovery, unavoidable errands, and nights that reduce the body's ability to reset before the next day. But if we ask how a heat warning is treated here, the answer is: not as a fixed temperature threshold. It is, rather, a locally defined trigger for the practical question of how residents can find enough opportunities to cool down across an entire day. Cities know this problem, the article tells us. Cooling centers, climate shelters, and related interventions are, of course, the most visible public metaphor for the act of lending protection to others. But if we ask how a list of such facilities has shown residents a workable daily strategy, the answer is: it has not shown it. It has, rather, catalogued it. We do not know from a map pin alone how people can move, where they can realistically stay, or how much between-stop exposure they must absorb along the way. The list eschews everyday route logic; there is no guidance there about the felt intervals of recovery, hesitation, and continued movement. This gap becomes visible at the scale of everyday mobility, the article tells us. But that mobility has no single-sensor interior. Cooling refuge is not a fixed thermal something - more like an ensemble of shaded, breathable, and socially usable conditions. It is being able to cool where, and when, one actually is. In one register, user-generated content on platforms such as Google Maps offers distributed descriptions of comfort, breeze, quiet, water, shade, and recovery that can be analyzed systematically to reveal perceived environmental benefits in urban space (Busch et al., 2024; Heikinheimo et al., 2020). (see Appendix Figure A1) In another, the article turns those comments into an auditable cooling index and embeds that index in a route-construction interface. The method does not claim to replace thermal measurement, but offers a low-cost conversion chain through which local experiential evidence can supplement formal

heat planning, while lightweight proxy checks against canopy, vegetation, and water proximity test whether the resulting scores stay directionally aligned with established cooling mechanisms (Bowler et al., 2010; Oke, 1987). More importantly, the study does not stop at mapping, but treats heat adaptation as a route-planning problem in which residents connect multiple cooling opportunities into workable daytime sequences instead of relying on a single destination.

Using Gelsenkirchen, where parks, river corridors, and post-industrial landscapes coexist, the paper proposes a low-cost method for identifying informal cooling nodes from local comment data. It extends heat-adaptation analysis from static point information to the intraday path level, emphasizing continuous exposure, recovery opportunities, and everyday access. In this way, local experience becomes interpretable and adjustable evidence for urban planning and community heat-resilience discussion.

PART 2 BACKGROUND & RELATED WORK

2.1 From the list cooling supply to the intraday heat adaptation path

Dealing with the urban heat island effect usually starts at the "point" level. When we shift our perspective from "facility supply" to "personal day", there will be a structural mismatch - most policy results are presented in the form of lists, points and coverage analysis, while people's experiences involve continuous exposure and continuous decision-making. The most common policy wording involves a single point or a single type of intervention, such as a summer shelter, a spray station or a cold roof or road surface. The assessment usually revolves around indicators such as "whether there are cooling measures", "what is the service radius" and "whether it covers vulnerable groups" (Adams et al., 2023; Gasparrini et al., 2015). These efforts are extremely necessary because they transform heat risks from an abstract climate issue into actionable issues related to infrastructure and service allocation, and also enable the discussion of "equity" to be concrete. Which communities are space closer to cooling facilities? Which groups can get better services? (Ebi and Hess, 2020; Harlan et al., 2013) (see Appendix Table A2)

Going to the summer resort center does not mean that you can "successfully escape the summer". The risk of heat stress does not only occur in the hottest two hours of the afternoon; it will persist with accumulated heat exposure and inability to recover, especially when the temperature is higher at night. Sleep and physiological reset ability will be weakened, and the heat resistance threshold of the human body will also be reduced the next day (Guo et al., 2024; Minor et al., 2022). This misalignment is more obvious in the context of the "hot night". If a person only stays somewhere for a short time, his health benefits may be limited. More importantly, people can repeatedly get enough opportunities to cool down during the day and reduce the high exposure area to an acceptable range (Gasparrini et al., 2015; Gronlund, 2014). That is to say, the core unit of thermal adaptation is usually not a "single location", but "the order and connection of multiple locations". (see Appendix Table A3)

Recent studies are increasingly linking thermal comfort with travel, which has given rise to the planning concept of "hot health route" for walking and cycling. For example, factors such as cold point, itinerary and cumulative heat load will be considered in the route recommendation, and the thermal risk will be transformed from a "heat island on the map" into an actionable problem of "what is the temperature along the way" (Neset et al., 2022). This kind of research reminds us that thermal adaptation is not only about macro planning or health advice; it can also be a practical daily navigation logic.

The focus here is on how to transform the information of scattered summer resorts into practical path guidance that residents can use on high-temperature warning days. Put more plainly, this article discusses how to help residents gradually transition from "knowing where it is cooler" to "knowing how to move, where to stay and which summer shelters are really available" within one day. (Hägerstrand, 1970; Neset et al., 2022).

These factors are usually not included in traditional thermal indicators, but they do directly affect the practical application, thus affecting the actual effect of the policy.

The article turns this level from a "soft factor" to an auditable methodological object. This means that whether a place can become a summer resort depends not only on thermodynamic factors, but also on "whether tourists are willing to stay", "whether tourists can stay quietly for a period of time" and "whether tourists feel comfortable and welcome". The article does not regard the experience as a decorative narrative, but transforms it into the input of indicators, and then transforms the indicators into the decision-making rules of itinerary planning. (see Appendix Table A4) (see Table 1)

Table 1. Heat-relief approaches and their planning objects

Approach family	**Typical evidence**	**Main planning object produced**	**What it does well**	**Where it breaks for day-long coping**
Cooling centers / climate shelters	Facility lists, service hours, catchment analysis	Points + coverage/equity metrics	Targets vulnerable groups; actionable siting and operations	A “pin” is not an itinerary; limited guidance on sequencing and between-stop exposure
Urban cooling interventions (misting, cool pavements, roofs)	Field measurements, intervention evaluations	Intervention performance metrics	Quantifies physical cooling effects	Does not translate into “where to go next” decisions for residents
UHI / land-surface proxies	Remote sensing, land cover, NDVI/canopy	Cooling capacity surfaces	Scales well; comparable across space	Weak on perceived comfort, safety, willingness to stay
Heat-aware route planning	Mode networks + thermal indices	Route recommendations, heat-load along paths	Treats heat as mobility risk	Often under-represents experiential cues and “stop” desirability

This study redefines thermal adaptation as the object of itinerary planning. Under the condition of high temperature warning, what people need is an executable "daily adaptation route", which consists of multiple informal shelters and is constrained by factors such as accessibility, willingness to stay and continuous exposure control. Now, the contribution points here are clearer. This is not to replace the "summer center" strategy, but to fill the long-neglected middle layer, that is, the transformation mechanism from "what points the city provides" to "how residents connect these points to form a day". The following method section will provide a more detailed

explanation. Here we will explain how to use user-generated content (UGC) to compress the "experience described as cool" into a "cooling index" and integrate it into the route construction task, so as to transform thermal adaptation from "information" to "choice" (Amorim-Maia et al., 2023; Neset et al., 2022). Also, this article also has a chapter diagram corresponding to the data.

2.2 Transforming local comments into supplementary experience evidence

The argument also benefits from being explicit about what comments tend to overrepresent and underrepresent. Leisure sites, waterfront promenades, and scenic parks are more likely to attract commentary than bus-transfer points, pharmacy stops, social-service waiting zones, or institutional forecourts where people may need relief most but seldom leave public reviews. So the index should be read as a situated usability proxy, not as a full account of heat refuge. The key term is dwell feasibility: whether a place can actually absorb a short recovery pause without immediately expelling the user through noise, crowding, lack of seating, or social discomfort. (see Appendix Figure A2)

User comments are regarded as evidence of thermal adaptation. Intuitively, this seems to treat subjective feelings as data. However, it is more accurate to record the content generated by users not the temperature itself, but the microclimate and livability experienced by people. Thermal risks in cities are usually manifested as "local differences". The movement of shadows on the street, the change of wind direction, and the radiation difference between the canopy and the hard road surface - all of which divide the city into many small but crucial comfort zones. Traditional thermal indicators and remote sensing layers can describe "where it may be colder" very well, but they cannot answer "why people stay there, whether they are willing to return again and again, and whether they feel safe and worth it".

The comment just fills this gap. They use everyday language to link "environmental characteristics" with "use scenarios", such as "there is a shade at 4 p. m. ", "it's comfortable to walk along the water", "windy, but there are not many seats", "it's quiet here and suitable for sitting and resting". These languages are not

thermodynamic measurements, but they point out the most difficult and operable behaviors and stay choices in terms of thermal adaptation.

This section must also explain in detail the risks related to user-generated content (UGC); otherwise, the reviewer will focus the criticism on the methodological part. The first type of risk is the difference between the platform and the sample: which users are more likely to comment, which areas are more likely to be commented, and when the comments are more likely to appear. Platform data is not equivalent to "public opinion" in the sense of census. Different platforms have different behavioral mechanisms and data structures, and this platform feature is a common limitation in user-generated data research (Boyd and Crawford, 2012; Kitchin, 2014). (see Appendix Table A5)

The second risk is language ambiguity and context drift. The same word may have different meanings in different cities, different seasons and different places; "cool" sometimes refers to the temperature, sometimes to the wind, sometimes to the shade, and sometimes even to "few people".

The third risk is ethics and recognizability: comments are public, but this does not mean that researchers can directly copy the text without any processing, especially when the combination of location and time may point to a specific individual. The mainstream suggestion is to follow the principles of "reasonable expectation" and minimal intervention to handle public user-generated content, anonymize it, avoid copying searchable original statements, and explain the source and scope of use of the data (Metcalf and Crawford, 2016).

2.3 From CES to informal heat refuges: Transform "experience" into actionable indicators while maintaining explainability (see Appendix Table A6)

In terms of heat adaptation, one of the most easily ignored aspects is "heat refuge", and heat refuge actually involves two different areas: half of it is the physical

world (radiation, evaporation, wind, canopy, water body), and the other half is the social world (whether people are willing to stay, can they sit down, whether they feel unsafe and whether Squeezed out). The core contradiction that this article aims to explore is how to build a real bridge between "summer resort" and "summer resort". If we only define "summer resort" as "places with low temperatures", we will get many seemingly correct but impractical answers. For example, the NDVI map shows a green space, but there is no shady place to rest in the afternoon; the riverbank is very windy, but the road is directly sunny, and the cost of walking is very high; the square is sprayed with water mist, but there are many people and noisy, which makes people unwilling to stay. On the other hand, if we only define "summer resort" as "a place that people think is comfortable", it is easy to slip into an unverifiable narrative.

The research tradition of the Cultural Ecosystem Service (CES) provides an extremely suitable language for solving this "dual world problem". The "sense of tranquility/recovery" of the same green space comes not only from environmental characteristics such as tree crowns, water and wind, but also from factors such as roads, facilities, management, congestion and user environment (Chan et al., 2012; Daniel et al., 2012). The key to CES is not to beautify this experience, but to regard it as a structured benefit, emphasizing that it usually co-produces (or co-produces) with biological and physical supplies. Applying this to heat adaptation can avoid the common misconception that "being described as cool" is directly equivalent to "actually colder". A more reliable way is to define a "summer resort" as a comprehensive benefit, including microclimate regulation (closer to "regulation service") and a sense of recovery, tranquility and comfortable habitability (closer to CES), which can be "realized" or "interrupted" through accessibility and living conditions (Fish et al., 2016; Haines-Young and Potschin, 2018). The refuge is not the result of a single variable, but a chain: potential cooling ability (ability) - perceptible cooling signals and atmosphere (availability) - actual stay and reuse (acceptance).

The reason the UGC-cooling index method needs this chain is not only to make it "more like a set of theories", but to make the subsequent index design easier to understand. Here, word frequency is weighted into a fraction, which may raise questions - what does this score represent? And this chain can give a clear answer:

this score is not temperature, nor remote sensing heat island effect or facility coverage, but "the availability and selectivity of a certain location as a shelter under heat warning conditions". Ideally, the glossary and coding system should be naturally divided into different levels, and different types of clues should be recorded separately. The first layer contains biological and physical clues (such as shade, water, breeze, etc.), the other layer is related to experience and benefits (such as calm, recovery, pleasure, etc.), and the third layer is related to the implementation conditions (such as seats, crowding, safety, opening hours, etc.). In this way, when explaining the cooling index, there will be no confusion about the mechanism: when a place has a high score, it can be clearly indicated that it is due to "strong physical clues", "strong experience clues" or "both are strong, but punished for congestion". This interpretability is not the rhetoric of writing, but the uncertainty of user-generated content is controlled within the scope of auditability. (see Appendix Table A7)

More importantly, the CES perspective naturally shifts the focus of these papers from "better maps" to "more operational planning objects". Because once it is recognized that shelters constitute a chain of "ability-affordability-utility", it is difficult to be entangled in the order of various elements. This is highly consistent with the "daily path and time limit" emphasized by time geography: human behavior does not follow the shortest path in abstract space, but follows a series of feasible sequences under the conditions of time budget, physical ability and accessibility (Hägerstrand, 1970). People's practical practice is to build a daily route by connecting multiple informal heat refuges, controlling the exposure time, arranging stopovers, and weighing under the constraints of reality. The "route construction task" applies the constraint logic usually used for travel research to thermal adaptation in the public domain, making "cooling down" a negotiable and shared route choice, rather than a hot and cold map.

2.4 Lightweight verification

This strategy of "using alternative indicators for comparison" is not lazy. It has almost become the default language for the study of the urban heat island effect. The reason is very simple: on-site measurement of microclimate is expensive, greatly affected by time and space, and it is difficult to cover the whole city. In contrast, remote sensing and open geographic data provide an extensible approximation method. Researchers often use NDVI, canopy coverage, land coverage, sky visibility factor (SVF), distance from water bodies, and even LCZ (local climate zone) categories as comparison or explanatory variables to discuss differences in potential cooling capacity and heat exposure (Bowler et al., 2010; Debbage and Shepherd, 2015; Oke, 1987). Of course, these proxyes are not the temperature itself, but they have a clear relationship with shading, evaporation, surface heat storage, radiation exchange and other mechanisms, so they are very suitable for taking on the role of "nailing the method back to the ground" (Bowler et al., 2010; Oke, 1987). The article positions the "proxy test" as a "strengthening tendon", which means that this article does not need to answer "how much has decreased", but needs to answer "Is the indicator test carried out here in the correct physical direction? " (see Appendix Figure A3)

PART 3 METHOD

3.1 Data sources and sampling

The UGC part mainly relies on Google Maps location comments because they are closely related to specific locations and contain the most basic structured metadata, such as timestamps and ratings, which makes them suitable for complete chain translation from text to indexes to routes (Goodchild, 2007; Hamstead et al., 2018; Heikinheimo et al., 2020). (see Appendix Table A8)

The data strategy of this study follows the principle of "low cost but verifiable", capturing daily thermal comfort clues through publicly visible user-generated content (UGC), while using open proxy data to provide minimal physical consistency checks. So, we have established a list of candidates with the jurisdiction of Gelsenkirchen as the boundary, focusing on the common categories of green spaces and public areas

where people can stay, including urban parks and community green spaces, riverbank trails and linear corridors, transformed industrial landscapes (such as mines), woodland or naturalized areas, and some public indoor spaces related to adapting to high temperatures (such as libraries/community facilities, if they clearly indicate "cool" in the comments). The sampling unit is divided into the location layer and the review layer. The goal of the location layer is not to cover all points of interest in the city, but to cover the main types of "potential informal informal heat refuges" and ensure the operability of subsequent spatialization and path calculation.

3.2 Coding scheme: from comment clauses to auditable cooling glossary

The coding goal of this study is not to "emotionally summarize" the comments, but to extract the repeated and location-related clues in the comments and include them in the subsequent index in an auditable way. To avoid treating the entire comment as a single label, we use the smallest semantic unit at the clause level: first, we split the comment into short sentences or clauses according to punctuation marks and conjunctions, and then perform multi-label coding for each clause. (see Appendix Table A9)

The coding framework adopts a three-layer structure, which corresponds to the "capacity-availability-absorption" chain proposed above: the first is the biological and physical clues (cooling clues), which record the perceptible elements that clearly point to the cooling mechanism in the annotation, such as: shade (shade/shade), water (fountain/river/water mist), breeze (wind /Ventilated/open-air corridor), cool surface (cool ground/cool waterside), etc.; followed by clues to describe the experience and benefits of staying and recovery ability (experience and benefits), such as: calm (quiet/noiseless), recovery (recovery/recovery/refrefering), comfortable stay, excellent scenery Beauty (landscape pleasure), etc.; the factors that cause "potential cooling" to become "unavailable" are the implementation conditions and obstacles (constraint conditions), such as crowding, no seats, sun exposure, inability to enter (construction/closure/unsafe), noise, etc. The advantage of the existence of these factors at the same time is that the cooling index can be mainly based on clues, but

can still be punished or explained by constraints, so as to avoid misjudging "very green but difficult to stay" places as high-availability shelters.

Aggregate clause-level multi-label coding to the location layer to obtain the normalized count and coverage of each location in the three layers of cues, benefits and constraints (for example: the proportion of SHADE mentioned in every 100 comments; Claus coverage with CALM/RESTORE; penalty density of constraints). The coded output is not directly equivalent to the final index. On the one hand, these intermediate indicators enter the composition of the Cooling Index as a "readable reason" for explaining the panel, so that both users and reviewers can trace where the index comes from.

3.3 Formulation of cooling index

The cooling index in this study is not a "measured temperature", but an auditable decision-making signal: it compresses the cooling clues, experience benefits and implementation conditions in the comments to the same scale, thus serving the itinerary planning problem, that is, "how the high temperature warning day is linked to the stopover point". Unlike traditional thermal risk assessments, the index targets availability and selectivity: given a local high-temperature alarm trigger, which locations are more likely to provide repeatable mitigation opportunities, and this judgment can be traced back to specific terms, weights and rules, rather than being covered by the aesthetic interface or black box models. (see Appendix Figure A4)

This layering is in line with the "potential-perceptible-realization" chain of ecosystem service cascading, avoiding the direct equivation of empirical vocabulary and physical mechanisms (Haines-Young and Potschin, 2018). In terms of implementation, the cooling index is calculated at the location level. For each location i, there is a set of cleaned comments r_i and coding results at its term level. The coded output is organized into cooling clues (such as SHADE/WATER/BREEZE, etc., which are direct indicators of cooling conditions); cooling benefits (such as

CALM/RESTORE, etc., which are related to recovery, tranquility and livability); feasibility modification factors (For example, overcrowding, insufficient seats, exposed areas, opening hours, etc., which are the implementation conditions used as penalties or constraints).

In this way, the indicator not only reflects "whether users repeatedly mention certain cooling conditions", but also avoids the deviation of a single extreme comment on the ranking. Similar robust processing methods are often used in content analysis and platform data research to control exposure differences and sparse sample deviations (Haklay, 2010; Kitchin, 2014). In order to reduce the noise caused by the uneven distribution of comments, this study adopts a structure that combines proportion and confidence discount. First, it converts various clues in a specific position into an appearance rate (the proportion of comments containing at least one coding clause). Then, use a confidence coefficient that increases with the increase in the number of comments to reduce the weight of the position with a smaller sample size. The index consists of the base clue income score (plus point factor), penalty score (deduction factor) and confidence factor (trust discount). The computable form used in this study is shown below. (see Appendix Table A10)

The location layer is based on the basic unit construction of urban space, which are "reachable, reachable and identifiable in comments". We have included parks, riverbanks/water edges, city squares, community pocket green spaces, post-industrial slag piles (piles) and their surrounding trails as candidate sites. For each location, we record: name, type, geometric type (point/face/line), representative coordinates (mass center or entry point), platform link/ID, administrative jurisdiction and accessibility annotation.

The selection of proxy variables used in lightweight verification follows the principle of "low cost, clear direction, and cross-city reusability": vegetation (NDVI or canopy coverage), water proximity, hard road surface ratio, etc. These variables are not equivalent to the temperature of the pedestrian scale, but they are obviously

related to the mechanism of shading, evaporation and surface heat storage, so they are suitable for directional consistency testing (Bowler et al., 2010; Oke, 1987).

3.4 Light verification: proxy reasonableness test

The light verification in this study is not meant to estimate causal effects or recreate on-site microclimate measurements. It works as a consistency check. Once comment language is turned into a cooling index and fed into route decisions, the least we need to know is whether the results still point in the same direction as the better-established physical principles in urban heat-island and blue-green infrastructure research.s a piece of methodological self-definition, the article is quite specific about what it is offering here: a method chain and an auditable rule system. Its standards, its claims, accordingly remain spare and untechnical. Where a stronger validation discourse would be heavy, this one travels light: verification is treated as a reasonableness check, which keeps the method within the actual scope of a methodology paper (Goodchild, 2007; Haklay, 2010). In short, the goal is convergence, not the assertion that the index equals temperature or that proxy variables can replace thermometers. (see Appendix Table A11) (see Appendix Table A12)

This writing method is clean-cut and, in the main, properly modest; it turns verification into a form of post-explanation. In this study, a set of the most basic expected directions is set in advance before the analysis. Under the condition that other factors remain unchanged, the cooling index should be positively correlated with the vegetation index (NDVI, canopy coverage), negatively correlated with the hard road surface index (water impermeability), and negatively correlated with the distance from the water body. At the beginning, it is equivalent to adding a "theoretical obstacle" to the index here. If the direction is reversed, you must go back to the codebook and weight table to explain which rule caused the mistranslation, instead of using the narrative to cover up the problem (Boyd and Crawford, 2012; Haklay, 2010). The intensity of the impact is not defined here, and "significant" is not the only criterion. This is because the user-generated content reflects perceptible and selectable evacuation clues, which will be affected by accessibility, congestion, seats,

order and safety and other factors; verification only needs indicators and agents to not conflict in the overall direction, thus proving that the translation chain will not systematically amplify language noise into physics. Unreasonable recommendation.

The relevant test answered the question of "whether they generally develop in the same direction", while the group comparison answered the question of "whether the high score position is really different from the low score position on the agent". In terms of implementation, I divide the groups into high groups and low groups according to the conential number of the cooling index. I compared the distribution differences of proxy indicators such as canopy/NDVI and water distance, and used the effect volume to represent the actual size of the difference (if it is expected that reuse in other locations can further enhance the persuasiveness, but will not significantly increase the workload, the repeated test can be carried out within the main location type to prove The clear direction relationship is not driven by a single type). (see Appendix Table A13) (see Table 2)

Table 2. Proxy plausibility checks for the Cooling Index

Dataset	Proxy	Spearman ρ	95% CI	p-value
Full set (N=85)	Tree canopy (%)	0.52	[0.33, 0.67]	<.001
Full set (N=85)	NDVI (mean buffer)	0.47	[0.26, 0.63]	<.001
Full set (N=85)	Distance to water (m)	-0.41	[-0.60, -0.18]	.002
Full set (N=85)	Imperviousness (%)	-0.36	[-0.56, -0.12]	.006
Stable subset (n=72)	Tree canopy (%)	0.49	[0.28, 0.66]	<.001
Stable subset (n=72)	NDVI (mean buffer)	0.44	[0.21, 0.62]	.001

When presenting the results, first of all, it is pointed out that the cooling index has a consistent directional relationship with the canopy/NDVI, distance from the water body, hard road surface and other indicators, thus supporting the rationality of the index as an available approximate signal; secondly, it is emphasized that the test cannot replace microclimate temperature measurement, nor can it be used for causal

inference; its purpose is only to provide low-cost rationality testing for the conversion chain of user-generated content to the index; finally, it is explained that stricter on-site measurement and cross-city verification belong to the follow-up extension work, but this does not affect the core contribution of this research, that is, "how to stably operate the auditable conversion chain under the background of planning". By adopting this method, verification can not only be "down-to-earth", but also avoid pushing this article to an overly heavy and unexpected heavy ecological empirical road.

3.5 Route decision-making logic under high temperature warning conditions

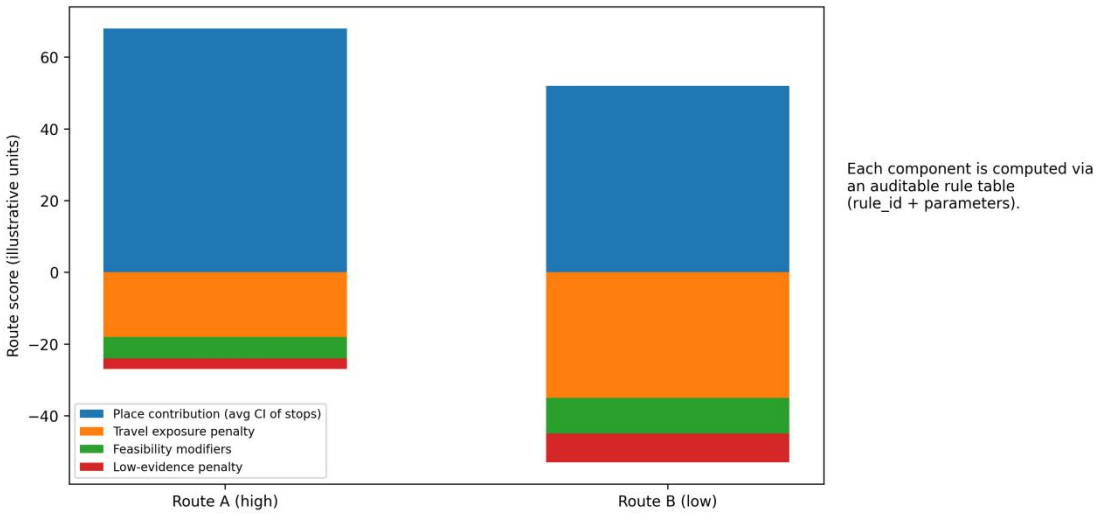
This study treats heat adaptation as an intraday itinerary problem rather than a static map problem. The interface translates place-level cooling evidence into route-level guidance by combining three components: place contribution, travel exposure, and feasibility modifiers. In practical terms, the score answers a local planning question: which sequence of stops is most likely to provide repeated recovery opportunities under heat-warning conditions, and which route segments remain weak points because access, exposure, or stay conditions break down? (see Appendix Table A14) (see Appendix Table A15)

Feasibility modifiers are the key link between thermal potential and lived usability. Public comments often describe benches, crowding, noise, safety, shade timing, and willingness to stay. The model therefore treats these conditions as auditable discounts rather than decorative experience language. A place may have strong cooling potential but still fail as a refuge if it is difficult to use, while an ordinary site may become important because it is quiet, connected, and genuinely inhabitable.

In this way, an important but real phenomenon may appear in the route planning. A location may have a good crown coverage/NDVI score, but it is excluded due to the "no stay" rule; instead, a place with general physical characteristics may become a

reasonable node in the route planning because it is "quiet, has a place to sit, is close to the water source and is willing to stay". This mechanism makes the route scoring truly a comprehensive result of "from experience to action", not just a repackaging of remote sensing maps. (see Appendix Figure A5) (see Figure 1)

Figure 1. Route score decomposition into place contribution, travel exposure penalty, and feasibility modifiers



PART 4 RESULT: CASE OUTPUT

This chapter takes the place layer and path task output of Gelsenkirchen as an example, and shows the observable results from "annotation language" to "position index" and finally to "path decision-making". Taking Gelsenkirchen as an example, this chapter explains how the method identifies informal cooling nodes in a real urban environment and further organizes them into hierarchical path outputs with local planning significance. The main purpose of the results is not to establish universal climate laws, but to prove whether local observation data can stably supplement existing thermal adaptation information and help identify cooling resources that can really be used in daily life, such as those that are easily accessible, connectable and available. (see Appendix Table A16)

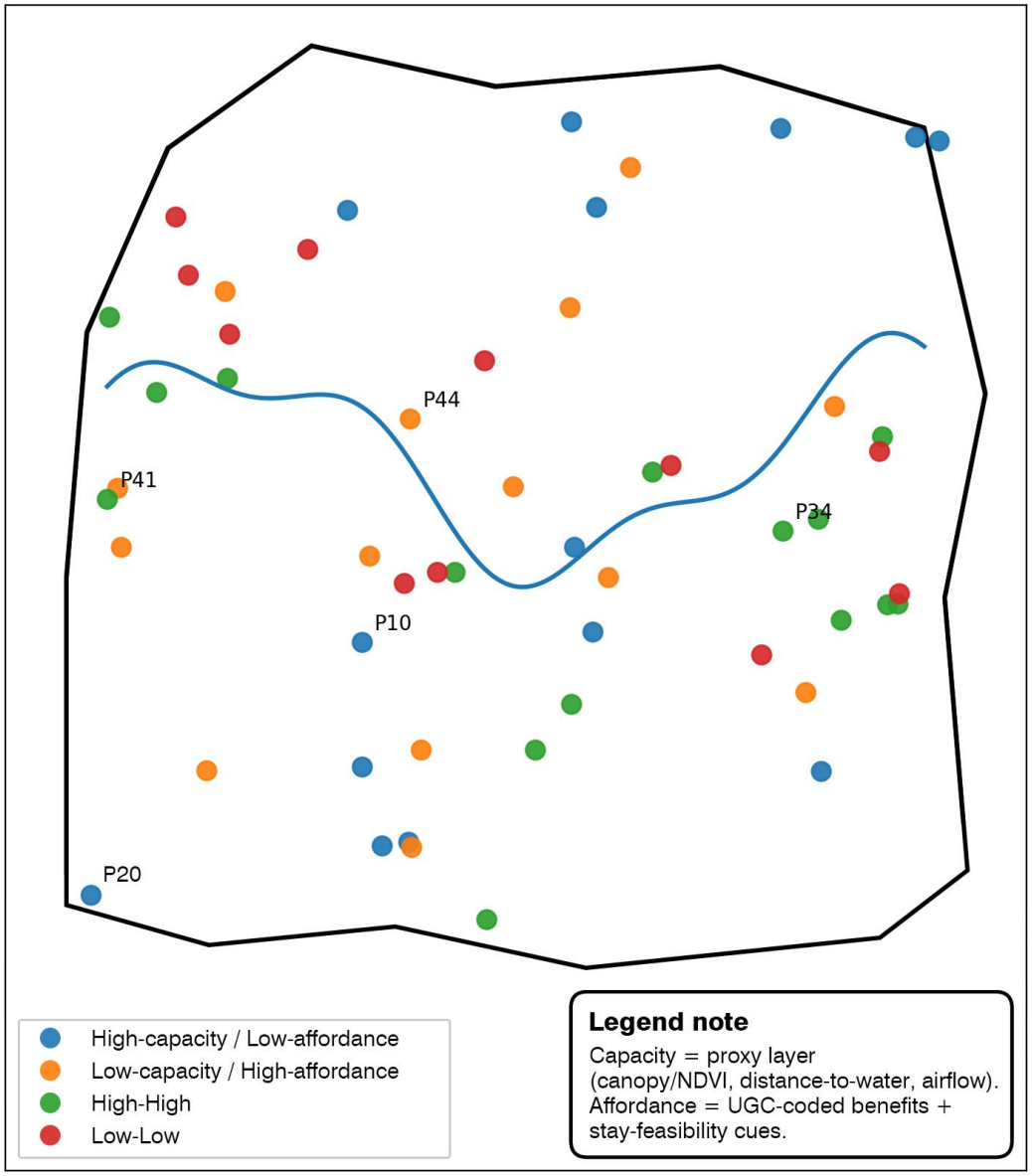
4.1 place-level results

The output of the position level is initially presented in the form of a calculable "position layer", and each position corresponds to a cooling index (0-100) and its components, as well as the number of comment evidence and uncertainty marks. Some types of locations have high scores in terms of clues (shady/waterscape/bree), but will be ignored due to factors such as overcrowding, lack of seats and noise; while other places may not stand out in other indicators, but they will become available nodes on the route because of language clues such as "willing to stay". This "unequality between capacity and availability" structure is exactly what this article wants to capture in the context of transferring the CES chain to informal heat refuges (Chan et al., 2012; Fish et al., 2016). According to the data, the average cooling index of 85 locations is roughly between 60 and 70, and the distribution is slightly right-deflected, indicating that there is a group of candidate shelters, which are unanimously identified as "eassible to recover/cooler" by comments. Still, a considerable number of middle and low score locations were reserved as control groups. More importantly, the distribution differences between different types of locations are not only determined by "green or non-green", but also present a comprehensive structure consistent with the theoretical framework of this article. (see Appendix Table A17)

So as to more directly support the previously proposed "capacity-availability-absorption" chain, this study also puts forward a series of "mismatched patterns" site selection classifications. The mismatch here is not a defect, but the most strategic output result. Locations with high crown/NDVI values, if they lack calm/restorative language characteristics, or the proportion of "unstoppable" modifiers is high, it belongs to the category of high capacity and low availability. At this time, what may not be needed is more tree planting, but seats, sunshade facilities, management rhythm and passages; instead, the lower generation Locations with rational but high language characteristics usually indicate that there are some miniature shelters that are ignored by traditional warming products, such as local shelters along the water, shady short stops or facilities with social comfort. This classification transforms the question of "why people don't feel cool in these places" into an output, comparable and

operable question, and is no longer limited to intuitive interpretation (Bowler et al., 2010; Oke, 1987). (see Figure 2)

Figure 2. Capacity–affordance mismatch map



Finally, the results at the location level must also solve the problem of uncertainty. The number of comments in some locations is relatively small, which are marked as "insufficient evidence" and marked with a gray outline or "insufficient evidence" label in the map and ranking output. There is no need to use complex Bayesian models for "correction" here. The more effective practice in the method

paper is to regard it as a visual constraint. These positions can be included in the path as exploratory nodes, but their contributions will trigger conservative penalties and be clearly marked in the interpretation chain, so as to avoid tools wrapping weak evidence into highly certain recommendations. As I have suggested, for reviewers this processing method itself is a manifestation of credibility, because the active management of user-generated content here is not being evaded but made examinable (Boyd and Crawford, 2012; Haklay, 2010). (see Appendix Table A18) Nor is that credibility exhausted by the index alone.

4.2 Route level mode

The most telling route results are the ones that a purely scenic or vegetation-focused method would be least likely to prize: mid-ranking nodes, modest in themselves, yet placed at just the right interval between exposed segments. The value of the paper lies in the forthright application of this route logic to real recovery. It is quite right, for example, to argue that a node with shade, seating, and a strategically timed position may matter more than a greener site that can be reached only by a long radiant crossing. In fact, the point could be pressed even further. Not only does the everyday success of a heat route depend on the environmental quality of each isolated stop; it depends on sequence and on the timing of pauses. (see Appendix Table A19)

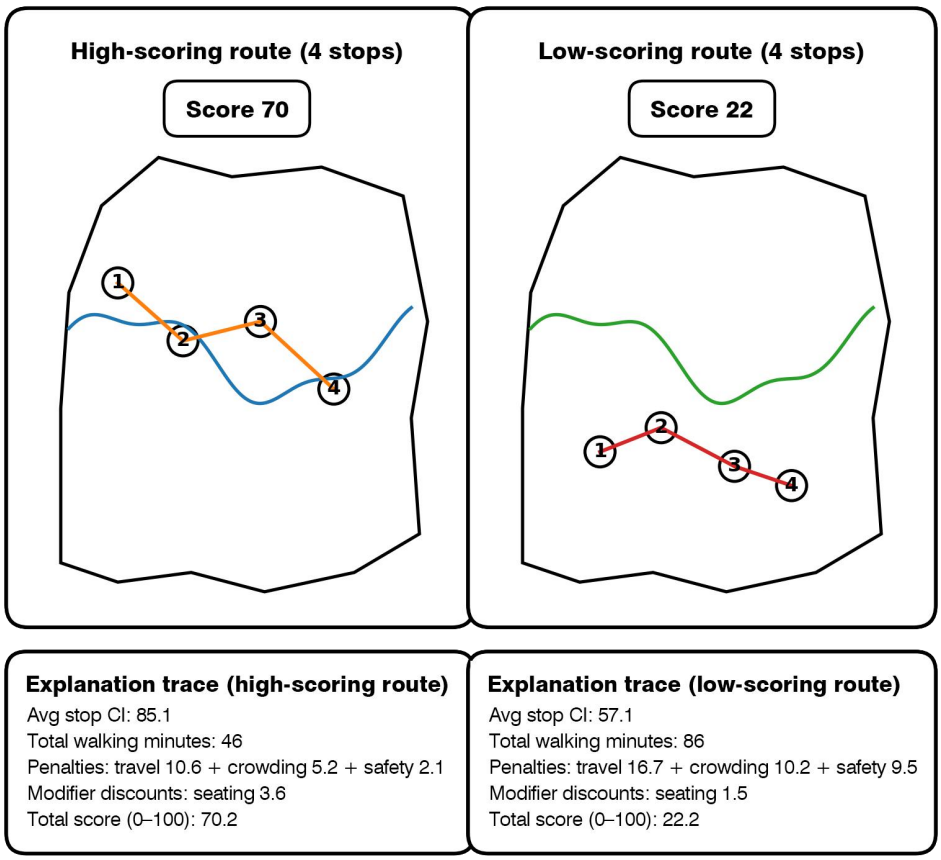
The results at the route level sort the locations in an executable order by day. Under the task condition of "3-4 stations, total walking time limit, accessibility constraints", N=1240 feasible routes were simulated. The average score of the route is 71, the standard deviation is about 10, the median is about 72, and about 58% of the routes have reached the "good scheme" threshold (for example, ≥70; this threshold can be localized, and the analysis is not limited to any absolute value). This distribution has two characteristics that are very beneficial to methodological papers. On the one hand, not all routes get high scores, indicating that the scoring system is distinguished; on the other hand, not only a few routes get high scores, indicating that the tool is not just "telling people not to go here", but can provide multiple feasible solutions that can be discussed under actual constraints. This makes the role of route

construction more similar to planning communication and public guidance than risk intimidation.

The road map layer best illustrates the argument that "the map is not a plan", because the route score often varies significantly due to the "gap" between the two points. The value of these results is that it redefined the concept of "avoiding hot weather to cool down" as "arranging the rhythm of recovery", which is the decision-making unit that actually occurs in urban life (Hägerstrand, 1970; Neset et al., 2022). (see Appendix Figure A6)

In the comparison chart, the two routes may choose high-score stations with similar scores. However, as long as one of the routes takes longer to walk, has less shade along the way, or must pass through a dense hard road surface area, its travel exposure penalty will significantly reduce the total score; conversely, even if a route contains a station with a medium score, as long as it acts as a "recovery window" in chronological order, and And the connecting section is relatively gentle, and its total score may still be higher. (see Figure 3)

Figure 3. Two example routes (high vs low) with explanation traces



The results at the route level can also be presented with the frequency of "anchor stops"", thus showing a type of urban structure that can be discussed. In this structure, some locations repeatedly appear in high-score routes, not because they are the strongest in proxy indicators, and more about they usually have both the characteristics of "docable" and "easy to connect", that is, they can provide both recovery windows and serve as connection nodes in the network. Such anchor stops are of direct significance to planning: they are "available" public cooling nodes, and they are also systematic bottlenecks that need to be maintained and protected. Once the anchor stop fails due to construction, closure or congestion, the entire feasible path collection will collapse. This vulnerability can be quantified through changes in path distribution. Method papers do not necessarily require complex scenario simulation. However, several "control scenarios for removing specific anchor stops" can be used as optional results to further demonstrate the sensitivity and interpretability of the tool in planning interventions. (see Appendix Figure A7) (see Table 3)

Table 3. Most frequent anchor stops in high-score routes

Anchor stop (name/ID)	Frequency in routes ≥70	Count (N routes)	Typical cue profile (SHADE/WATER/CALM)	Common modifier risks
Nordsternpark (P01)	44%	546 / 1, 240	SHADE high | WATER med | CALM high	Crowding (weekends); limited seating at peak
Emscher Riverside Walk (R03)	39%	484 / 1, 240	SHADE med | WATER high | CALM med	Path congestion; low night safety in segments
Rheinelbe Heap Viewpoint (H07)	34%	422 / 1, 240	SHADE low | WATER low | CALM med	Heat exposure; wind discomfort; steep access
Consol Park Pocket Greens (G05)	29%	360 / 1, 240	SHADE high | WATER low | CALM high	Seating scarcity; perceived safety after dusk
Heßler Park Grove (P12)	26%	322 / 1, 240	SHADE high | WATER low | CALM high	Crowding near playground; noise spillover
Canal Edge Rest Spot (W09)	23%	285 / 1, 240	SHADE med | WATER high | CALM high	Slippery edges; safety signage needed
Science Park Lawn (F02)	21%	260 / 1, 240	SHADE med | WATER low | CALM med	Open pavement heat; glare; low evening comfort
Zoo-side Shade Loop (A14)	19%	236 / 1, 240	SHADE high | WATER low | CALM med	Queue congestion; animal-area noise peaks
Rungenberg Heap Ridge (H11)	17%	211 / 1, 240	SHADE low | WATER low | CALM med	Heat + exposure; long walking penalty
Community Garden Courtyard (G18)	15%	186 / 1, 240	SHADE high | WATER med | CALM high	Limited access hours; gate/maintenance constraints

PART 5 DISCUSSION

5.1 Contribution

In many urban environments, official thermal adaptation products are often better at describing spatial distribution, but not very good at solving the questions that

individuals and communities actually need to answer, such as where I need to go today to get recovery time, whether I can park there, and how I should connect multiple nodes into a viable route. Also, the assembly chain deliberately avoids a common trap, that is, treating user-generated content (UGC) as "subjective noise" or treating alternative indicators as "objective truth". This study uses the chain of "capacity-availability-use" to redefine "informal heat refuges", which also incorporates biophysical potential and social implementation conditions, thus explaining why some areas with high NDVI values are not used as informal heat refuges, while some seemingly ordinary nodes repeatedly appear in the search. In the narrative of seeking shade (Boyd and Crawford, 2012; Goodchild, 2007). (see Appendix Figure A8)

5.3 Prejudice and Ethics

The biggest risk of using platform comments is not "its subjectivity", but "it may be mistaken for representativeness". The reviewer group may differ from the overall population of the city in terms of age, language, travel ability, digital literacy and space usage habits. The recommendation mechanism of the platform also affects which locations are more likely to be commented on and which narratives are more likely to be seen. So, using user-generated content (UGC) as the premise of perception means treating it as structured evidence, not polling. Specifically, this study first controls the deviation through audit. What we disclose is the coded count and proportion, the summary data at the location level and the uncertainty mark, not the authoritative truth of the original text as "evidence is the truth". We clearly mark the position with a low number of comments (apply conservative penalties in the route score) to avoid packaging weak evidence into highly determined recommendations. (see Appendix Table A20 and Appendix Figure A9) (see Appendix Table A21 and Appendix Table A22)

On the other hand, deviation control is achieved through comparison. The proxy rationality check cannot eliminate deviations, but it can reduce the situation of "language noise is obviously abnormal at the level of physical common sense"; while mismatch classification presents deviation and governance problems separately, so

that the discussion can focus on "why potential cooling measures are not used" instead of arguing about "whose experience is more real" (Gould et al., 2020; Haklay, 2010). (see Appendix Table A23)

5.3 Planning significance

From a planning perspective, the main value of this study is that it translates scattered cooling-site information into a local discussion framework. But these route-level outputs, I should argue, are plainly useful. Indeed, the whole issue becomes less crudely put once one asks which nodes actually provide recovery, which connections remain exposed, and where modest interventions such as shade, seating, access, or opening hours would most improve everyday heat resilience. For the problem of refuge assumes too easily that cooling capacity is the thing that matters, while the terms of use remain fixed. (see Appendix Figure A10) Only on the basis of this false assumption would one miss the justice dimension in the results. Heat adaptation does not reduce, in the sense that thermal ranking does, to environmental performance alone. But the governance picture does shift and deepen. For instance, locations with high cooling capacity but low availability point to resources that exist without being usable at the right times or in the right ways. Small and ordinary sites, by contrast, can become crucial refuges when they remain accessible, connected, and trustworthy. This mismatch allows local actors to discuss heat adaptation as a problem of distribution and everyday usability rather than thermal performance alone.

5.4 Limitations and next steps

A further limit is strategic. This method is good at surfacing route-level usability, but it should not be mistaken for a substitute for heat sensor networks, public-health data, or targeted engagement with groups who are least likely to post online. Its virtues, not the least of which are discipline and realism, are narrower. It offers a local translation chain that makes informal refuge evidence discussable within planning while remaining open about where the evidence is thin, biased, or provisional. (see Appendix Table A24) The limitations of the study are structural and should be read

that way. The cooling index is not a temperature measure; the proxy checks are not causal estimates; public comments do not represent the whole city. What the article does make explicit, rather, is the conversion chain itself: auditable, local, and open to adjustment. Future work can extend the model with on-site measurements, cross-city comparison, and finer-grained mobility data without changing its core planning logic. (see Appendix Figure A11 and Appendix Table A25)

PART 6 CONCLUSION

The central claim here is that adaptation to extreme heat should not be reduced to finding a cooler point on a map. It also concerns whether residents can repeatedly reach, use, and connect recovery spaces across an entire day. By combining public comments, light proxy checks, and route-level reasoning, the study offers a low-cost and explainable method for supporting local heat planning and everyday climate justice. (see Appendix Figure A13 and Appendix Table A26)

Ethics and data availability: The study analyzes publicly accessible user-generated content in anonymized, aggregated form; no identifiable user text is reproduced; derived data are available from the corresponding author on reasonable request. (see Appendix Figure A12 and Appendix Figure A14)

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